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Sorting by colour in the food industry

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5.1 Introduction

Food has colour, and good food is often a different colour from bad food. This in itself is not a very remarkable statement but its implications for the food industry are very significant. The human perception of colour has proved very effective in the determination of quality in many foods, and the sorting of food particles using the human eye and hand has been, and still is, widely practised. Machines designed to replace the hand sorter obviously need to duplicate some of the functions of the human eye, brain and hand. Automatic colour sorting is often referred to as 'electronic sorting', as it was the advent of electronics that made possible the duplication of the eye and brain functions.

Experiments with automatic sorting equipment started immediately after World War II and colour sorting machines gained early acceptance in the food industry. Demand for automatic sorting machines continues to increase owing to the escalating costs of hand sorting coupled with the higher quality requirements being imposed on food producers. There is also an increasing realisation of the importance of sorting in the reduction of health hazards arising from contaminated food, whether due to the food itself or the presence of foreign bodies such as stones or wood, and the avoidance of associated product liability claims which are escalating dramatically in both number and value.

5.2 What is a sorting machine?

Colour sorters generally consist of four principal systems, as shown in Fig. 5.1:

1. A feed system, which presents food particles to the subsequent systems in a controlled manner. Figure 5.1 shows a feed system which involves a conveyor belt to align the particles and ensure that they pass through the machine at a constant velocity. However, feed systems can also be based upon inclined chutes, contra-rotating rollers and different combinations of these features.
2. An inspection system, which measures the reflectivity of each particle. The inspection components are housed in an optical chamber, through which or past

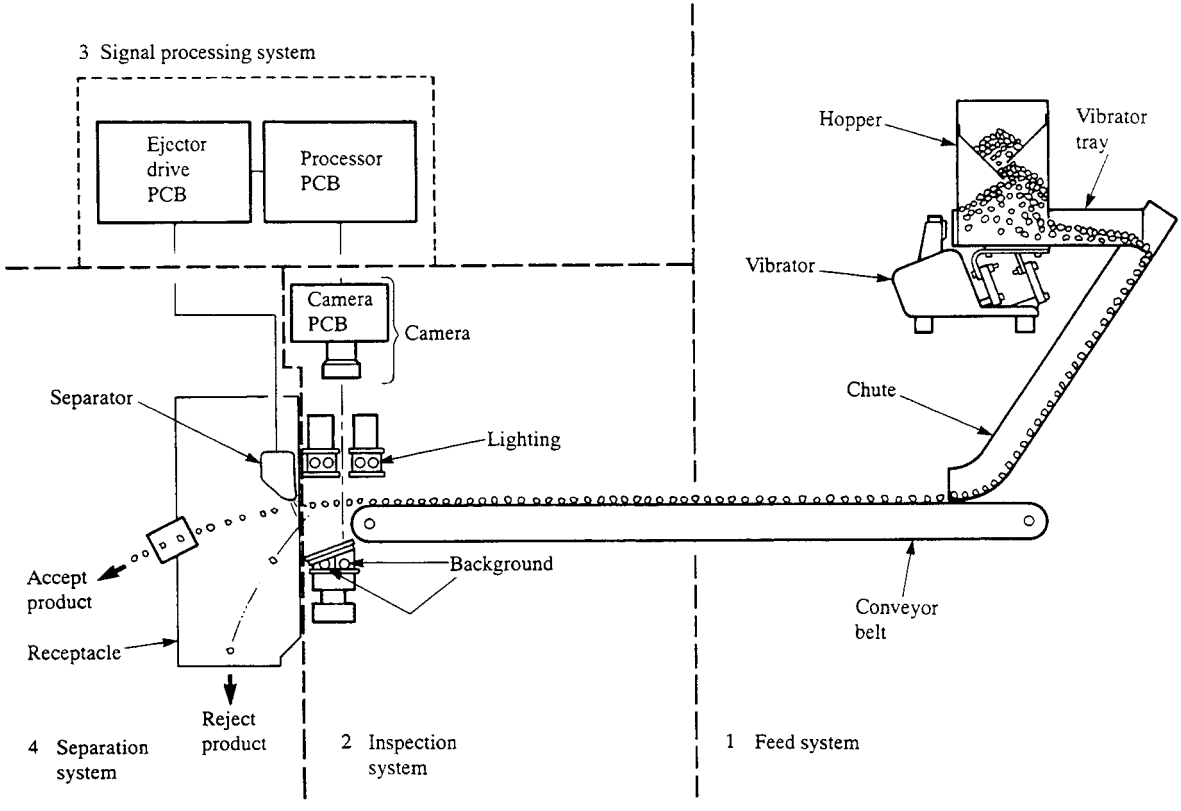


Fig. 5.1 Sorting machine systems.

which the particles travel. Particles do not come into contact with any part of the optical chamber and are separated from it by glass screens or windows. The optical chamber contains one or more lens and detector units, depending on the number of directions from which the particles are viewed. Lamp units, designed to provide even and consistent illumination of particles, are also usually contained within the optical chamber.

3. A signal processing system, in which electrical signals from the detectors are amplified and fed to decision-making circuitry which classifies particles as either 'accept' or 'reject'.
4. A separation system, capable of physically separating particles classified as reject from those classified as accept. Usually, separation takes place while the particles are in free fall; accept particles are allowed to continue along their normal trajectory, and rejects are deflected from it into a receptacle. Deflection is usually achieved by emitting short bursts of compressed air through nozzles aimed directly at the rejects, although large or heavy particles may require some sort of piston-operated device to deflect the rejects.

The size, cost and complexity of sorting machines varies depending on the size range of particles to be handled, the throughput requirement and the complexity of optical measurement. Machines are employed in sorting particles as small as mustard seeds; however, rice grains are among the smallest particles to be sorted on a large commercial scale. At the other end of the size range are vegetables and fruit such as potatoes and peaches.

Seeds are usually sorted on a single- or double-chute machine at a throughput of 60 to 600 kg/hour. A higher throughput can be achieved on a multi-chute or a conveyor belt machine; rice at 16 tonnes/hour (320,000 objects/sec) and peas at up to 16 tonnes/hour are typical examples. The simplest type of optical measurement is taken from one direction and is of a single wavelength band. More complicated optical measurements involve two or more wavebands and may be made from several directions around the particle. The products that can be handled by today's automatic sorting machines include seeds, coffee, rice, breakfast cereals, nuts and pulses; fresh, frozen and dehydrated vegetables; cherries and tomatoes; and biscuits and confectionery. Foreign material such as stones, sticks and organic matter can be removed, as well as particles with defects such as discoloration and damaged skin. Figures 5.2 and 5.3 show two typical sorting machines.

5.3 Assessment of food particles for colour sorting

5.3.1 Spectrophotometry

To determine whether a particulate food product is suitable for colour sorting, and which type of sorting machine and optical configuration is most suitable, samples of both acceptable and unacceptable produce must be measured and assessed in the laboratory. Actually the term 'colour sorting' is a misnomer, possibly arising from the effect on the product of sorting. The criterion used when assessing food particles for sorting, and what the sorting machine measures when it inspects the product, is spectral reflectivity at particular wavelengths, rather than the colour as a whole. Figures 5.4 and 5.5 illustrate typical spectral curves obtained from a product. Intensity varies from black (0) to white (100), and wavelength covers the visible spectrum (400 to 700 nm) and extends into the



Fig. 5.2 Sortex 90000 is a gravity chute machine, designed for bulk sorting of dry products such as coffee, or rice. The machine is just under 2 m in height, 1.5 m wide by 2.5 m deep.

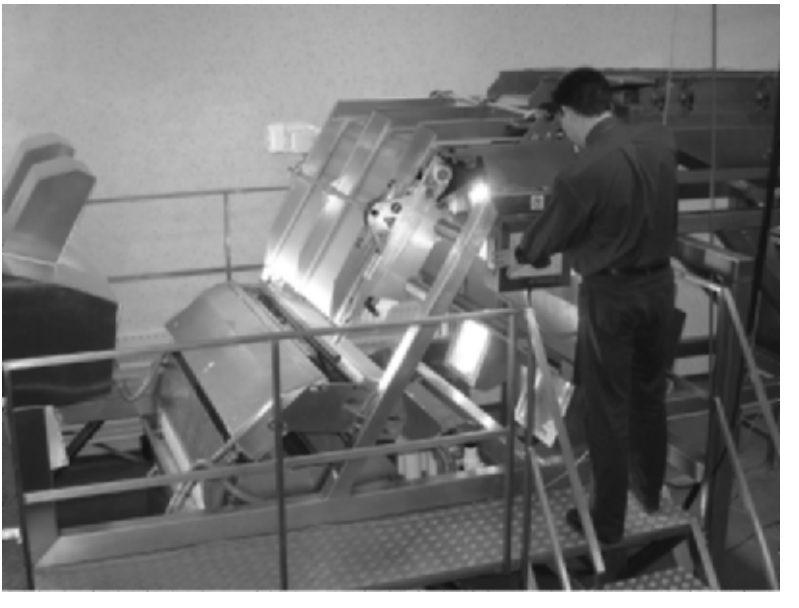


Fig. 5.3 Sortex 'Niagara' is a broad belt sorting machine, designed for simultaneous colour and shape sorting, of frozen or fresh vegetables. Niagara is just over 1.5 m in height, 2 m wide by 3 m long.

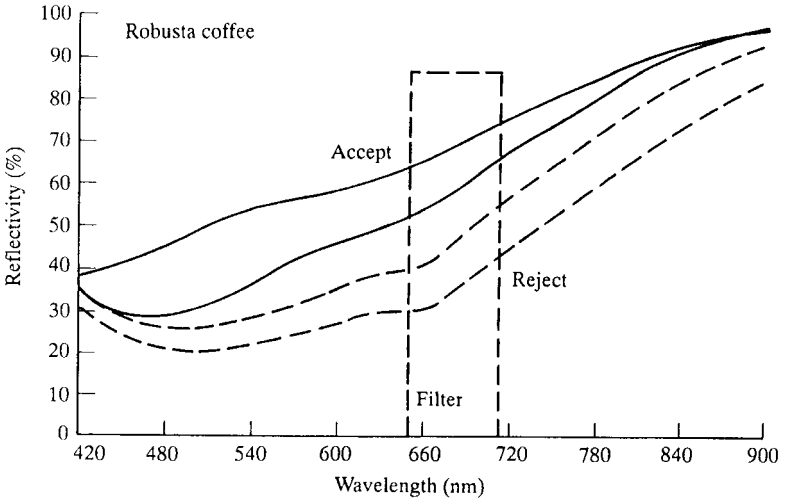


Fig. 5.4 Spectral curves – monochromatic sorting; response curves for — 'accept' samples, – – – 'reject' samples.

near infra-red. Ideally there will be a part of the spectrum where intensity values for all acceptable produce are either higher or lower than values for all unacceptable material; if so, this part of the spectrum can be used as a basis for sorting. In practice the spectrum is normally restricted to a narrow waveband in the sorting machine by the use of optical filters.

Conventional spectrophotometry in the food industry involves the measurement of carefully prepared surfaces under standard optical conditions and illumination. However, practical industrial sorting machines must deal with naturally occurring surfaces viewed under non-ideal light conditions. To obtain the best possible result for a given product, special computerised reflection spectrophotometers have been developed which enable the appropriate characteristics of the naturally occurring surfaces to be measured. Diffuse

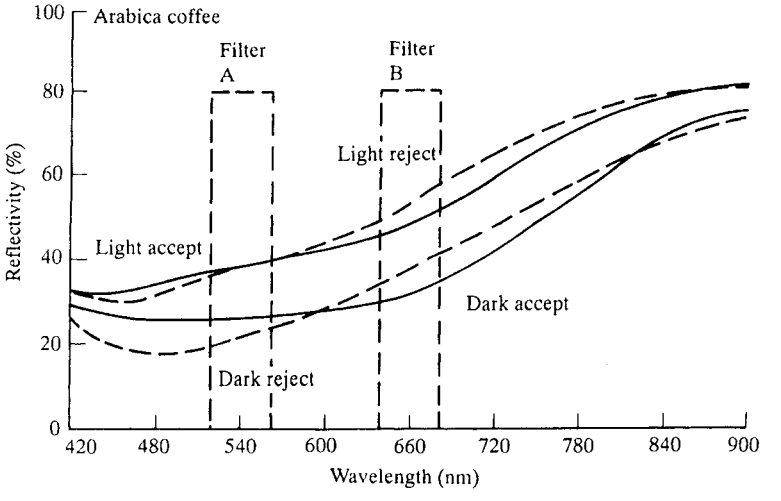


Fig. 5.5 Spectral curves – bichromatic sorting; response curves for — 'accept' samples, – – – 'reject' samples

spherical broadband lighting is used to illuminate uniformly the particle under test. The reflected light is then passed through a computer controlled scanning monochromator, which splits the light into its constituent wavelengths. The output is measured using a suitable detector and the data are fed into the computer. When the equipment is appropriately calibrated, the results can be plotted showing the variation of reflectivity with wavelength for both acceptable and defective product.

5.3.2 Monochromatic sorting

Sorting that uses the form of optical measurement described in the preceding section is known as monochromatic sorting. It is so called because it is based on the measurement of reflectivity in a single isolated band of wavelengths. For it to be effective, there must be a distinct difference in reflectivity within the selected waveband between all the acceptable particles and all the reject particles (see Fig. 5.4). Removing dark, rotten peanuts and removing black speck from rice are typical applications of monochromatic sorting.

5.3.3 Bichromatic sorting

Unfortunately, it is not always possible to find a single section of the spectrum where the intensity levels of accept and reject material are clearly separated, and in these cases it is necessary to resort to a more complicated procedure. This is known as bichromatic sorting, and involves measurement in two different wavelength bands. Figure 5.5 shows two sets of spectral response curves obtained from green arabica coffee. One set of curves (solid curves) represents the lightest and darkest of acceptable beans, the other (dashed curves) represents the lightest and darkest of discoloured beans; clearly, there is no wavelength where there is a complete separation of the two sets of curves. However, it can be seen that between 510 nm and 660 nm the difference in the gradients of the two sets of curves is at its greatest. If measurement *A* is taken at 540 nm and measurement *B* is taken at 660 nm, and the ratio *A*:*B* is calculated, then this will give an unambiguous separation of the two particle types. (In principle, measurement *A* could be taken at 510 nm, but this would yield a lower signal intensity.) Bichromatic sorting involves duplication of optical and detection components, the addition of light-splitting devices and more complex signal processing. Therefore it is used only when a simple monochromatic measurement is unable to give a satisfactory result.

5.3.4 Dual monochromatic sorting

A third type of measurement, similar to bichromatic in that two wavebands are observed but without employing the ratio of the two measurements, is known as dual monochromatic. As its name implies, this system sorts monochromatically in each of two separate wavebands. This type of sorting is used when it is necessary to reject two types of defect or, for instance, defects and foreign material, each of which exhibit different spectral characteristics.

An example of dual monochromatic sorting is found with white beans; maize is rejected by detecting blue reflectance, and white stones are rejected using near infra-red. Some practical separations require both monochromatic and bichromatic decisions for the successful removal of all defects. Therefore automatic sorting machines are available which are capable of making both types of measurement simultaneously.

5.3.5 Trichromatic sorting

Bichromatic sorting techniques are clearly extendable to trichromatic measurement. The information gained from the third band-pass filter is often used for detection of gross defects such as the presence of foreign material, including glass, stones, thistle heads, caterpillars and mice! Trichromatic sorting almost always uses either green, red and infra-red band pass wavelengths, or less typically for the food industry, red, green and blue.

Trichromatic sorting allows objects to be sorted according to their size or shape, by suitable modifications to the sorting algorithm. In this way objects of the same colour, but differing shapes can be separated. For example, pea pod can be distinguished from peas and green stalks, or green caterpillars from green beans. Under or over sized objects along with mis-shaped objects, with holes or cracks can also be detected and effectively removed. The Sortex Niagara machine is capable of simultaneously sorting for both colour and shape at a rate of 40,000 objects/second.

5.3.6 Fluorescence techniques

Of course, not all bad food is a different colour from good food. It has been found that certain non-visible defects (bacteria for example) fluoresce when irradiated with long-wave ultraviolet light (350 nm), and this property may be used as a basis for sorting. This technique was originally developed for removing ‘stinkers’ from green arabica coffee beans, but has found applications in sorting peanuts, almonds and cranberries.

5.3.7 Infra-red techniques

Over the last decade, the wavelength range used by sorting machines has been extended from the visible, further into the infra-red region. Here, both water absorption and other chemical effects play an important part in determining the reflectivity characteristics of food particles. Bichromatic infra-red machines are proving particularly effective in removing shell fragments from a variety of tree nuts.

5.3.8 Optical sorting with lasers

Incorporating the use of lasers into bulk food sorting is a technique that is still in its relative infancy. A laser beam is used to illuminate the product and the degree of reflected light is affected by the amount of laser light that is either scattered from the surface, or diffused within an object. Since the laser produces narrow beams of coherent light at a single wavelength, there is no need to use optical band pass filters. However, the principal difficulty with this technique is in maintaining the high capacity demanded by the bulk sorting industry, in conjunction with the necessary resolution to detect defects accurately. The linear scan rate of the laser across the width of the view and the velocity of the product determine the vertical resolution. To date, laser scanning is limited to approximately 2000 scans per second. For product travelling at 4 m/s, the vertical resolution is therefore of the order of 2 mm. By comparison, line scan CCD technology can offer a resolution of 0.3mm. There are additional problems associated with the drop in illumination intensity and therefore, signal-to-noise levels that result by fanning out the laser beam across a line of sight. It can be quite a design challenge to scan a laser reliably and mechanically by say, a rotating polygon mirror, in the hostile temperature and debris ridden environments that are often encountered in food processing plants.

To a limited extent, some successful sub-surface and texture inspection can be carried out on certain soft fruits and berries with laser light. In fact, the technique has already been commercially deployed for some product areas in the food industry, notably for dried fruits like raisons, or certain vegetables, nuts and tobacco. There is certainly scope for further study and possible wider exploitation of the technique.

5.4 The optical inspection system

5.4.1 Introduction

At any given wavelength, the characteristics of the electrical signal from the detector will depend on the particle reflectivity and size, the light source, the properties of any filters fitted and the nature of the detector itself. The range of wavelengths measured by the machine is defined by the choice of these last three elements. Traditionally the light source and detector in a given machine were fixed, but the filters could be changed to accommodate different produce. In modern equipment it is often possible to fit lighting with different spectral characteristics and even to change the type of detector, as well as easily exchange filters.

Once the spectral feature or features which are to be used as a basis for separation have been identified, the relevant wavelength bands must be isolated by selection of appropriate filters and illumination. A primary objective of selecting filters and lighting is to obtain the maximum possible signal-to-noise ratio from the detector at the required wavelengths, and the minimum possible signal at all other wavelengths.

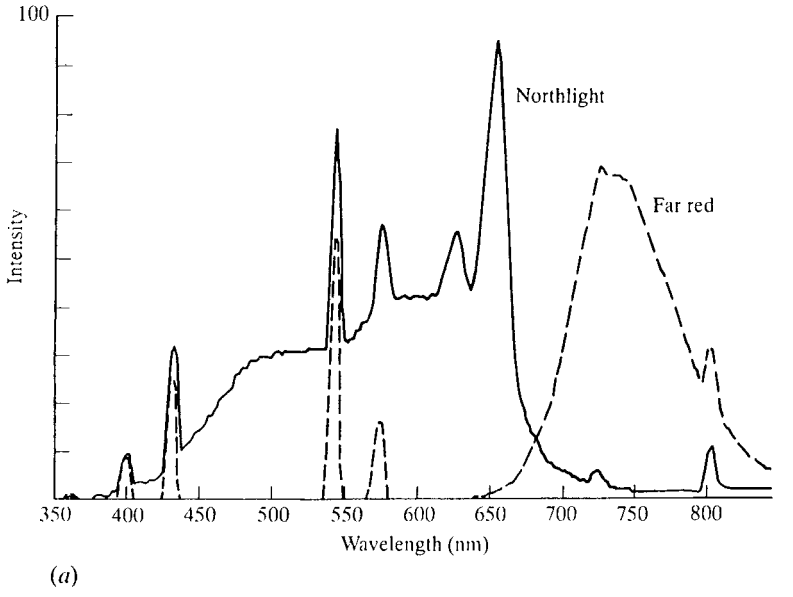
5.4.2 Illumination

When dealing with irregularly shaped particles, uniform diffuse illumination is necessary to minimise the occurrence of highlights and shadows, as these would obviously detract from the measurement of true surface reflectivity. Ideally, at the point of measurement, the particle should be surrounded by a spherical surface of uniform brightness; in practice, however, a number of factors preclude the attainment of this ideal. Firstly, if the inspection chamber is designed so that the product particles pass through it, there must be provision for entry and exit ports. Secondly, the position of the optical components will result in areas of different brightness compared with the main chamber wall. Thirdly, the use of light sources of finite size leads to non-uniform illumination.

A further problem that can arise, even with a perfect diffuse illumination sphere, is that of specular reflection. If a particle with a diffuse reflective surface is placed in such a sphere then its true colour will be observed. However, if the particle surface is not diffuse, specular reflection will occur, giving highlights which do not exhibit the true colour of the surface. Clearly the highlights can adversely affect the optical system and hence result in the incorrect classification of a particle.

In practice, illumination is provided by either fluorescent tubes or incandescent filament bulbs, or a combination of both. A number of lamps are arranged to give as uniform a distribution of light as possible; with incandescent lamps, screens are often positioned to diffuse the high-intensity point of light emitted by the filament.

Fluorescent tubes can be produced with different spectral characteristics, extending from the ultraviolet to the far red, as shown in Fig. 5.6. The advantages of the fluorescent tube are its cool operation, long life and diffuse light. Its disadvantages are low red emission and the need for a special power supply to prevent flicker.



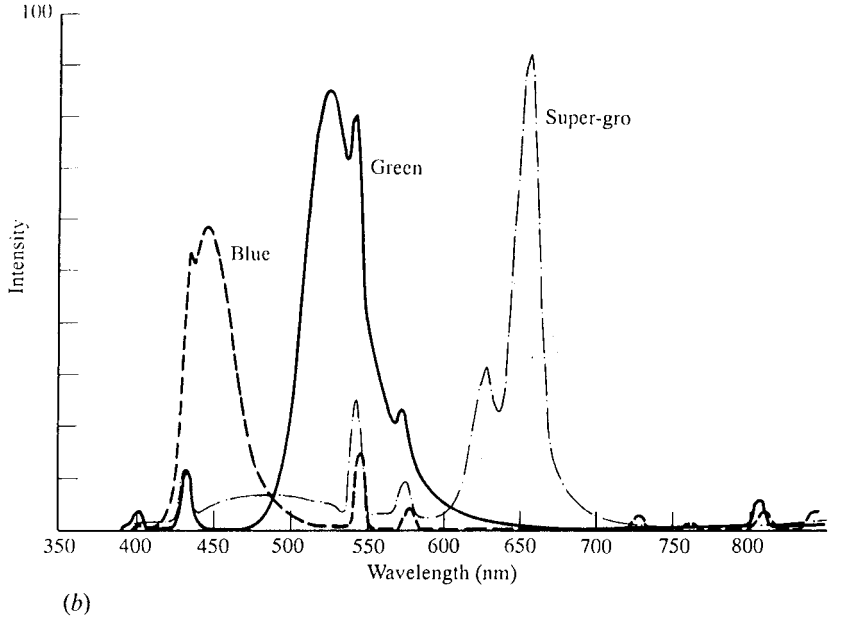


Fig. 5.6 Spectral characteristics of fluorescent tubes.

Incandescent lamps inherently have a wide spectral range from blue to the near infra-red, as shown in Fig. 5.7. These lamps have the advantage of wide spectral emission and DC operation, but suffer from being point sources which dissipate large amounts of heat. In general, the use of fluorescent tubes is favoured except in cases where a deep red or

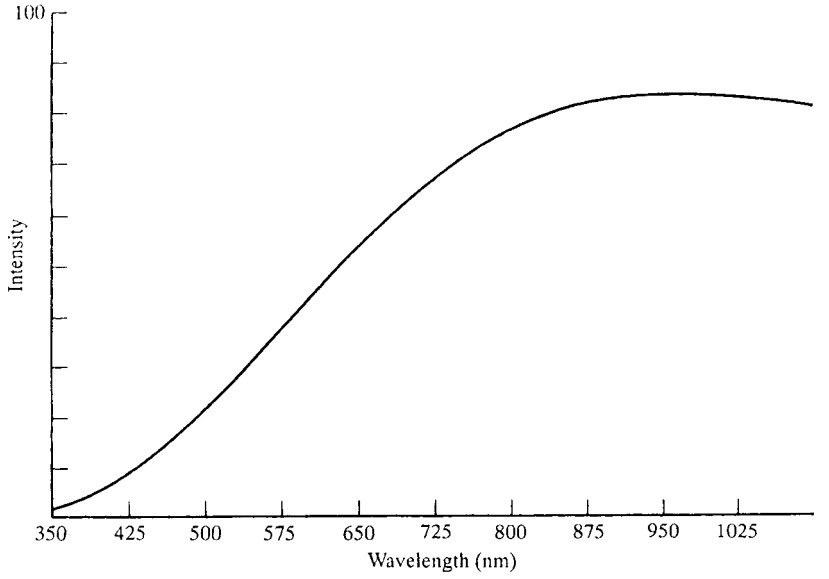


Fig. 5.7 Spectral characteristics of an incandescent lamp.

near infra-red measurement is required. The wider spectral range required for bichromatic machines necessitates the use of incandescent lamps.

5.4.3 Background and aperture

Having arranged for the particle to be illuminated in the optimum manner, it is necessary to consider the best optical arrangements for obtaining the necessary data. The simplest form of inspection system views the particles against an illuminated background and through a small aperture. The brightness of the background is adjusted so that the optical system measures the same average value with or without product. This is known as a 'matched' background because it matches the average brightness of product, including any defects.

The advantage of matching the background is that measurement of reflectivity is independent of particle size. For example, consider the case of a stream of particles containing rejects which are darker than accepts. If a matched background is used then whenever a defect passes across the aperture a decrease in signal amplitude will result, but a light particle will cause an increase in amplitude; thus an unambiguous decision can be made by the electronics. If, on the other hand, a background lighter than all particles were to be used, then all particles would give a decrease in signal; in particular, small dark defects could give signals identical to those of large light particles and hence the two could not be separated. Note that the brightness of the light reflected from a particle through the aperture is the product of the size of the particle and its reflectivity, including any area of discoloration. The background may be either an opaque surface reflecting light from suitably located lamps, or a translucent surface transmitting light from rear-mounted lamps or LEDs.

The aperture takes the form of a rectangular slit, as shown in Fig. 5.8. The width of the slit must be sufficient to allow for scatter in the trajectories of the particles and for the range of particle sizes. The height of the slit is kept to a minimum, consistent with

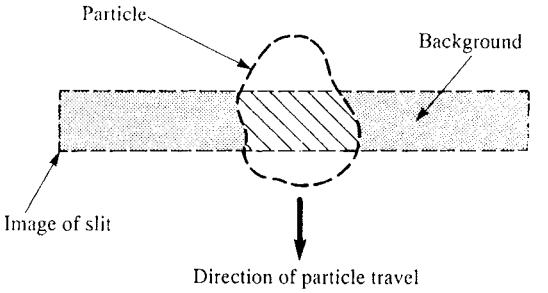


Fig. 5.8 Viewing aperture.

sufficient signal, in order to give maximum resolution and for accurate timing of the delay between detection of a defect and rejection of a particle.

5.4.4 Optical filters

A wide range of coloured glass and interference optical filters are available, making it a simple matter to select suitable filters. Three basic classes of filter are used:

- low pass, transmitting only below a certain wavelength
- high pass, transmitting only above a certain wavelength
- bandpass, transmitting only within a band of wavelengths.

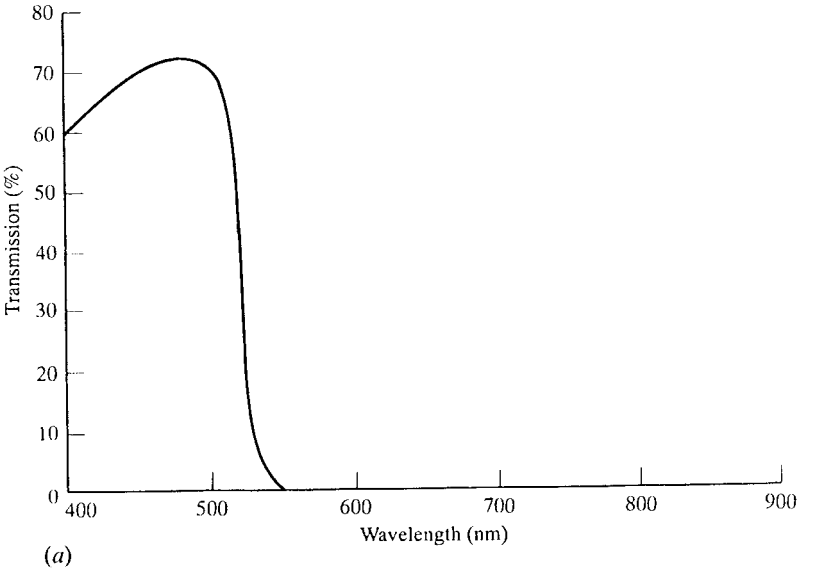
Figure 5.9 illustrates the response of typical filters used in sorting machines.

5.4.5 Detectors

For a number of years the photomultiplier tube was the best detector of visible radiation. It has a good signal-to-noise ratio which enables detection of low light levels, and a satisfactory blue response. However, it has the disadvantages of fragile mechanical construction, limited life, high operating voltage and poor deep red and near infra-red response. Following photomultiplier tube technology, the optical sorting industry moved on to use solid-state photodiodes as detectors. The photodiode has the advantages of comparative cheapness, mechanical robustness and an indefinite life. However, it has a poor blue response when compared with the photomultiplier.

Modern optical sorting machines typically use high-speed line scan CCD (Charge Coupled Device) technology. The silicon CCD offers the advantages of high spatial resolution, good quantum efficiency, high sensitivity and broad band response (400 to 1000nm). Despite being an analogue sensor, the output of a CCD is readily convertible to a digital form. This allows exploitation of state-of-the-art, low-noise, high-speed digital processors for subsequent signal processing. Unfortunately, CCDs still suffer from relatively poor response in the blue wavelengths (400 to 500 nm).

Other detector materials are often utilised in order to extend the detection range into the infra-red domain. Germanium and other materials are used for detection of foreign matter, up to approximately 1700nm. Historically, single photodiodes were used, however, Germanium linear arrays are commonplace and generally used in combination with silicon CCDs.



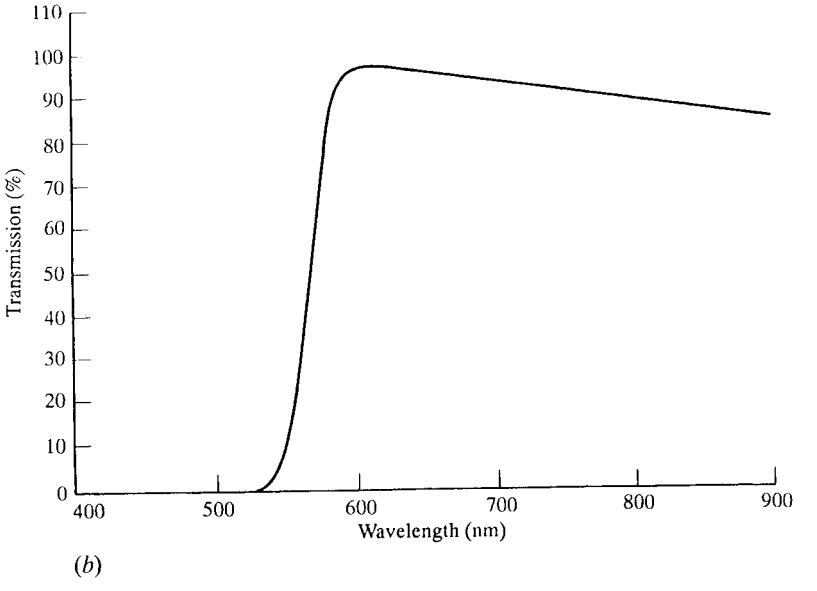


Fig. 5.9 (a) Low-pass filter (b) high-pass filter.

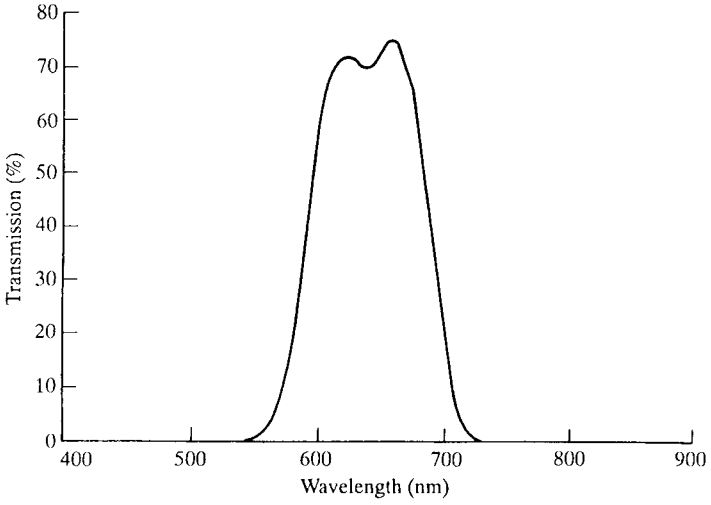


Fig. 5.9 continued. (c) bandpass filter.

5.5 Completing the sorting system

5.5.1 Feed

A product feeding system in a sorting machine has to provide three functions:

1. Metering, to ensure that the optimum number of particles per unit time is presented to the inspection system.
2. Acceleration to a constant presentation speed. The time taken for particles to travel from the inspection point to the separation point must be constant in order for the operation of the separation device to be accurately timed. Presentation speeds can be as high as 4 m/s and the delay between detection and separation lies between 0.5 and 100 ms.
3. Alignment, to ensure a controlled trajectory through the inspection and separation points.

In practice, the metering requirement is met by a vibrating tray feeder mounted below the output of a holding hopper. A variety of systems have been developed for providing the acceleration and alignment requirements, and the following are commonly used:

- inclined chute
- narrow grooved belt
- contra-rotating rollers
- flat belt single layer
- inclined belt single layer.

The first three methods above channel and separate the particles, allowing an all-round view of their surface. The flat belt system presents the particles in a single layer, restricting the view but enabling a much higher throughput of product to be achieved. Throughput is increased in channel feed systems by providing two or more channels.

With conveyor belt feed systems the inspection components may be mounted above the belt, with the belt serving as the background, or after the end of the belt so that the

product is viewed in free fall as in channel systems. The latter arrangement also enables the product to be viewed from multiple positions.

5.5.2 Separation

The usual form of separating device is the high-speed solenoid or piezoelectric valve which releases short bursts of compressed air through a nozzle. These devices exhibit the essential features of rapid action, reliability and mechanical strength; the fastest (a Sortex patented piezoelectric design) operates at a frequency of 1 kHz, firing a pulse of air for 1 to 3ms. Ejectors operate at input pressures between 200 to 550 kPa, depending on the size of the product. It is usual to arrange for the separation point to be outside the inspection area so as to minimise the effect on the optics, because the action of the air blast on a reject could cause dust particles and skin fragments to be blown around. The necessary time delay between inspection and separation is generated by the electronic circuits. Accurate timing of the delay time to coincide the ejector air blast with that of the object to be ejected as it falls in front of the ejector, relies on the product having constant velocity. In practice the variation in product velocity is about 5%. The trajectory of each particle also becomes harder to predict, the greater the distance between the viewing point and the ejection point. It can become a major design challenge to position the optical inspection and ejection system as close to the foot of the chute as possible.

The operational lifetime of the ejectors must be in the region of at least a billion or more cycles. Food processing is usually a 24 hour-a-day, all-year-round operation. Operators cannot afford to regularly shut down a machine for even a few minutes to replace faulty ejectors. Under these circumstances, machine reliability and stability of operation are critical. For certain large or heavy objects a solenoid valve may be used to control a pneumatically operated flap or plunger to deflect the reject. Specialised ejectors for product pulp remove rejects by suction and are mounted above a flat belt, downstream from the inspection unit.

5.5.3 Cleaning and dust extraction

The application of optical techniques in a dusty, dirty or wet industrial environment poses major difficulties for designers. Considerable effort has gone into designing sorting machines suitable for use in such conditions. To protect the optical components from dirt or moisture, they are separated and sealed from the inspection point through which product passes by a glass window. The position of this window in the optical path should be such that any small particles, which may settle on the surface, are out of focus and therefore cause the minimum of disturbance to the optical signal. Nevertheless, it is essential that this window is kept as clear as possible and a number of facilities to achieve this may be provided on the machine.

Firstly, the product being fed to the machine should be as dry and dust free as possible. However, the action of storing it in a hopper and feeding it on a vibrating tray may well create some dust and therefore a dust extraction nozzle may be fitted at the end of the vibrating tray. In the case of a chute feed, the top of the chute may be perforated so that air can be passed through the product stream to remove dust particles. Secondly, the actual window can be cleaned by means of compressed air jets which (a) provide a continuous curtain of air to prevent particles settling on the surface and/or (b) a periodic high-pressure blast which removes any particles that have managed to reach the window.

Thirdly, a pneumatically actuated blade or brush may periodically wipe the window. In more advanced machines this may be combined with an air blowdown facility.

Finally, any dirt created by the action of the ejector blast on the particles, may be drawn away from the window area by the provision of a dust extraction nozzle below the optical box. Similar precautions may be taken where a wet product is to be sorted, with water jets and wiper blades being fitted to the optical window.

5.5.4 Electronic systems

The electronic systems in sorting machines have progressed from the simple analogue circuits of the early machines to the sophisticated digital microprocessor based circuits found in the present generation of machines. The digital microprocessor machine exhibits two important advantages over its predecessors in that it is very much easier to set up and that it will maintain a consistent level of performance without the need for periodic operator intervention. Digital electronics are not subject to performance drift as a consequence of thermal or environmental change, in the same way that analogue electronics are affected. The signal-to-noise level is also greatly improved by using digital electronics, along with ease of subsequent data processing and transportation. Most of the setting up of the sorting parameters can be done by the machine itself including in some cases the ability of the machine to 'learn' the differences between good and bad product. However, the operator is always given the opportunity to fine-tune the final result.

If required the machine will track the average colour of the product so that, even though the product colour may change with time, the machine will continue to remove only the predefined abnormal particles. Alternatively the machine may be provided with a calibration plate which is periodically placed in the optical view so that the machine is able to correct for any measurement drift that has occurred. These facilities result in a much more stable performance when compared with that of machines that require periodic human attention.

Once a machine has been set up for a particular product then all the settings can be stored in memory. This can be repeated for a number of different products and then, at a later time, the machine can be made ready to sort any of these products simply by recalling the appropriate settings from the memory. As the machine has a memory capability this can be utilised to provide useful information, such as the number of rejects that have occurred in a certain time, to the operator and where appropriate to the plant monitoring computer. Information can also be provided to assist with preventative maintenance.

As a consequence of the greatly increased signal processing available, the sorting decision can be based on other parameters as well as reflectivity. For example, the minimum size of the discoloration necessary for a particle to be classed as defect can be defined. As well as providing signals to external monitoring devices, the machines can be controlled via a computer link from a remote position.

User interfaces

Microprocessor control and visual display devices have greatly simplified the procedure for setting up and operating sorting machines, as well as enabling more comprehensive fault information to be displayed to the user. A typical machine will have either a keypad and a display unit, or more commonly a touch-screen user interface (Fig. 5.10) which will enable the operator to set up and control the machine by means of a series of menus.



Fig. 5.10 Niagara touch screen user interface.

Nowadays, it is becoming common practice for the user interface to employ touch-screen technology. In addition the display unit will provide the operator with information regarding the settings of the machine while it is sorting, together with details of any faults that may occur.

Mapping techniques

The conventional bichromatic sorting machine using two wavelength bands, say green and red, made a decision based on the ratio of the two signals as well as the intensity of the individual signals. The situation can be represented by a two-dimensional map as shown in Fig. 5.11(a). Clearly, the defined accept region is triangular in shape whilst the actual region is not and therefore it is impossible to set the machine without some accept or reject product being on the wrong side of the lines. However, it is also possible to define the boundary between accept and reject to match the true shape of the actual product colour distribution as shown in Fig. 5.11(b). This method enables the microprocessor machine to remove a far greater range of defects, more accurately, without the penalty of removing a large amount of accept product. Obviously, these techniques can be extended into three dimensions for trichromatic colour sorting.

5.6 Future trends: computer vision systems

Computer vision systems are being used increasingly in general manufacturing, but the demands of the food industry are generally much greater. At present, there is only a limited range of computer vision equipment available for use in the food industry. However, in the future this is likely to change. Two factors are limiting the rate at which computer vision systems are being introduced to the food industry:

1. The data-processing rates required in a high-speed food sorting machine are very much higher than those in a similar inspection machine for manufactured items.

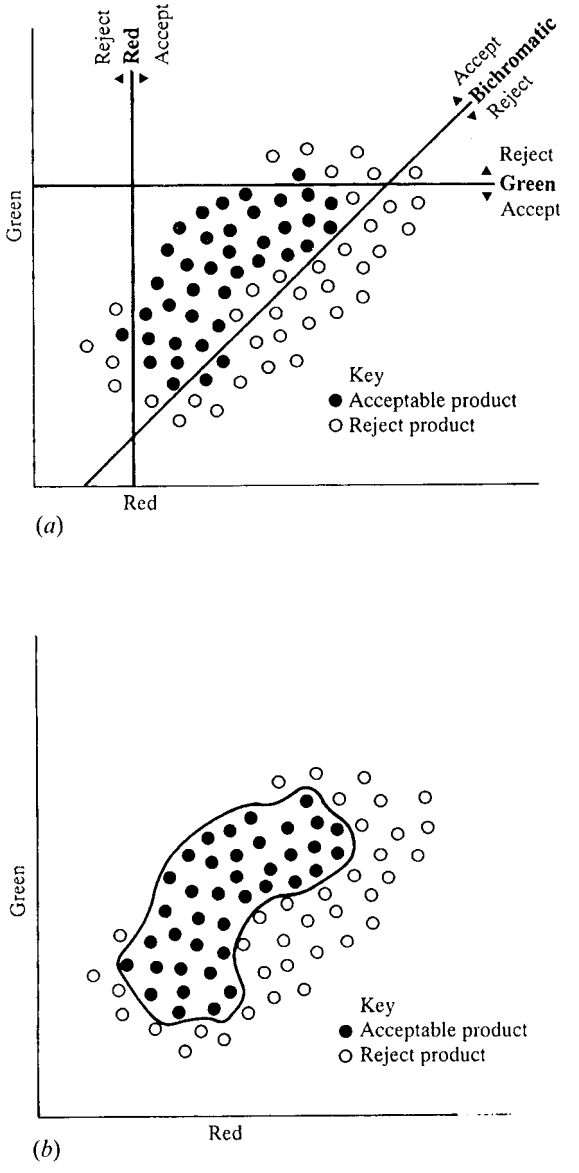


Fig. 5.11 (a) Conventional map (b) microprocessor map.

2. The development of improved material handling and separation systems is not keeping pace with the dramatic advances being made in computer technology.

The computerised sorting system of the future may comprise a TV camera and a computer with a large capacity for information storage. Some form of interface would connect the computer to a series of robot graders and sorters. Items to be inspected and sorted would pass along a belt where they would be viewed by the camera. The computer would identify the individual items and could assess them according to several criteria, including colour, shape and size. Devices would then be actuated to separate the items into size and quality grades.

A computer vision system potentially offers many benefits over a conventional colour sorter. The ability to sort food on the basis of several different criteria would be a primary advantage. Flexibility is another distinct advantage of such a system. A machine, which is busy sorting potatoes, could easily be reprogrammed to examine oranges, or indeed anything else that could pass along the conveyor. Different products could be accommodated by simple software changes instead of major adaptations to hardware. However, for the immediate future, the most likely application of advances in computer technology is the improvement in performance of the present generation of sorting machines.

5.7 Using a colour sorter

There is a common misconception that a colour sorter can remove all defects from a given batch of product. In practice this is impossible. A colour sorter will reduce the concentration of defective product, but it can never be 100 per cent effective. All colour sorters are bound to remove some acceptable product and fail to remove some of the defective product. There are a number of possible causes for this limitation. In some cases, the physical size or the colour difference of the individual defect may be too small for the machine to be able to see. The defective item will then be allowed to pass through the machine as if it were unblemished. In other cases, the machine may detect a defect

Table 5.1 Typical machine performance figures

Product	Machine	Purpose	Throughput (tonnes/hour) per machine	Typical efficiency (removal of defects)
Whole green beans	Niagara – Trichromatic colour sorter, 1m wide belt	Remove attached stems and blemish	8	Shape 80% Colour 60–70%
Green coffee	90003Bi – 48 channel, gravity chute bichromatic sorter	Remove defective beans and foreign material	3 to 6	99.5% (input contamination 6%)
Parboiled rice	90004 mono – 128 channel, 4 gravity chute, monochromatic sorting machine	Remove spotted, discoloured rice and foreign material (stones, glass, paddy, etc.)	5 to 10	Black 90% Brown 90% Rojo's 80% Yellow 70% Bites 60%
Frozen peas	Niagara – Trichromatic colour sorter, 1m wide belt	Remove foreign material, pea pod, sticks, etc. by colour and shape.	10 to 16	Foreign material 98% Blemish 70–80%
Mixed vegetables	Niagara – Trichromatic colour sorter, 1m wide belt	Remove foreign material	6	80–90%
Frozen lingonberries	Niagara – Trichromatic colour sorter, 1m wide belt	Remove green, white, pink and blue berries	4	80–90%

and remove the item, only for it to bounce back into the accept stream after it has been ejected.

Machines can be adjusted to optimise their performance. However, changing a given parameter may have some disadvantages in addition to the intended benefits. There is normally a compromise between achieving a high sorting efficiency and the amount of good material that is rejected. Sensitivity is one of the principal parameters that the operator can change. Increasing the sensitivity will result in the machine rejecting more defective material but a great proportion of good product will also be rejected as the sensitivity threshold approaches the average product colour. There are limits to throughput within which a colour sorter should work. If the product flow is increased above the upper limit, or decreased below the lower limit, optimum performance cannot be obtained. Within these limits a general rule can be applied. Increasing the flow of product through the machine will result in more good product being lost.

The colour sorter will normally achieve its best performance if the product to be sorted has been properly cleaned and size-graded prior to sorting. Size-grading product prior to sorting will allow a more precise setting of the ejection system and will allow better accept quality without increasing the amount of good product rejected. Proper cleaning of the product prior to sorting will reduce the build up of dust on optical surfaces and should improve performance between periodic cleaning of the machine. Table 5.1 illustrates some typical performance figures for a variety of products sorted on different machines. The throughputs are quoted in ranges, since the throughput increases as the level of input contamination decreases.

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